Journal of Product & Brand Management

A cross-national validation of the consumer-based brand equity scale

Isabel Buil Leslie de Chernatony Eva Martínez

A cross-national validation of the consumer-based brand equity scale

Isabel Buil
Department of Economy and Business Studies, University of Zaragoza, Zaragoza, Spain

Leslie de Chernatony
Birmingham Business School, The University of Birmingham, Birmingham, UK, and

Eva Martínez
Department of Economy and Business Studies, University of Zaragoza, Zaragoza, Spain

Abstract

Purpose – This study seeks to investigate the measurement invariance of the consumer-based brand equity scale across two samples of UK and Spanish consumers.

Design/methodology/approach – Brand equity was conceptualised as a multi-dimensional concept consisting of brand awareness, perceived quality, brand associations and brand loyalty. To test the brand equity scale cross-nationally a survey was undertaken in the UK and Spain. Measurement invariance was assessed using multi-group confirmatory factor analysis.

Findings – The brand equity scale was invariant across the two countries. Results show that the consumer-based brand equity scale has similar dimensionality and factor structure across countries. In addition, consumers respond to the items of brand equity in the same way, which allows meaningful comparison of scores.

Research limitations/implications – Future research could examine the cross-national generalisability of the brand equity scale using other countries' products and services.

Practical implications – Given that the brand equity scale is invariant across countries, researchers and international marketing managers can use this instrument to measure and manage brand equity across countries. This is suitable for testing theoretical and conceptual relationships in different national settings and allows managers to design and implement efficient international brand strategies.

Originality/value – The study contributes to the scarce literature testing the cross-national applicability of consumer-based brand equity. Furthermore, the research enhances consumer-based brand equity measurement by using a non-student sample and including a different type of brand associations and multi-item measures for all the brand equity dimensions.

Keywords Brand equity, Research, United Kingdom, Spain

Paper type Research paper

An executive summary for managers and executive readers can be found at the end of this article.

Introduction

Brand equity is one of the key topics in marketing in recent years. A strong brand with positive brand equity has several advantages such as higher margins, brand extension opportunities, more powerful communication effectiveness and higher consumer preferences and purchase intentions (Keller, 1993; Rangaswamy *et al.*, 1993; Cobb-Walgren *et al.*, 1995). As a significant asset (Ambler, 2003), research has focused on understanding how to build, measure, and manage brand equity (Keller, 1993, 2007).

Measuring brand equity is important due to its strategic value guiding marketing strategy, aiding tactical decisions and providing a basis for assessing brand extendibility (Ailawadi *et al.*, 2003). Thus, it has become essential for brand managers to understand how to measure brand equity (Ambler, 2003). In addition, the growing number of brands competing in international consumer markets necessitates developing valid and reliable brand equity measures that are generalisable across different countries.

International research needs standardised constructs and valid scales equivalent for different countries (Mullen, 1995; Steenkamp and Baumgartner, 1998; Knight *et al.*, 2003). Reliable assessment of cross-national measures is of fundamental interest to international companies since these measures influence the precision and quality of strategic decisions (Parameswaran and Yaprak, 1987). Assessing measurement invariance is a key issue for the successful study of any concept (Steenkamp and Baumgartner, 1998; Vandenberg and Lance, 2000). However, while a failure to establish cross-national equivalence may threaten the validity

The current issue and full text archive of this journal is available at **www.emeraldinsight.com/1061-0421.htm**

Journal of Product & Brand Management
17/6 (2008) 384–392

[ISSN 1061-0421]
[DOI 10.1108/10610420810904121]

The authors would like to thank the following sources for their financial help: CICYT (Ref: SEJ2005-02315) and the projects "GENERES" (Ref. S-09) and "PM026/2006" from the Government of Aragon. Isabel Buil wishes to express her gratitude for the financial help received from the "Caja de Ahorros de la Inmaculada" and DGA (CONSI + D).

of conclusions, constructs operationalised by researchers and marketing practitioners tend not to be subjected to rigorous testing (Durvasula *et al.*, 1993; Mavondo *et al.*, 2003). Often constructs and measurement instruments are transferred from one context to another, without examining the applicability across countries and/or cultures.

Despite the considerable interest in the concept of brand equity and its measurement, there have been few attempts at its cross-national validation. To the best of our knowledge, in the literature rooted in the cognitive psychology paradigm, only the works of Yoo and Donthu (2001, 2002) to date have explored the cross-cultural validation of the brand equity scale.

Addressing this gap, our research puts forward a consumer-based brand equity scale. We investigate and validate its cross-national applicability in two countries, the UK and Spain. Our objectives are: first, to establish whether brand equity can be conceptualised in the same way across countries; and second, to explore whether difference scores on the items can be meaningfully compared among the countries. Therefore, configural and metric equivalence will be analysed. In addition, this study addresses some of the limitations regarding consumer-based brand equity scales.

The article opens by reviewing the literature on brand equity conceptualisation and measurement. A briefly overview of the measurement invariance issue in cross-national research is presented. The methodology employed is explained followed by the analysis of empirical results. Finally, a discussion of the findings is presented and the managerial implications are drawn.

Literature review

Two main frameworks conceptualise brand equity. Keller's (1993) conceptualisation focuses on brand knowledge and involves two components: brand awareness and brand image. By contrast, Aaker (1991, p. 15) provides one of the most generally accepted and comprehensive definitions of brand equity:

> A set of brand assets and liabilities linked to a brand, its name and symbol that add to or subtract from the value provided by a product or service to a firm and/or to that firm's customers.

We draw on four of Aaker's five core brand equity dimensions: brand awareness, perceived quality, brand associations and brand loyalty. The fifth of Aaker's dimension, other proprietary brand assets, is normally omitted in brand equity research since it is not directly related to consumers.

The first dimension, brand awareness, refers to "the ability of a potential buyer to recognise or recall that a brand is a member of a certain product category" (Aaker, 1991, p. 61). This construct is related to the strength of a brand's presence in consumers' minds and is usually measured through brand recognition and recall under different circumstances (Keller, 1993; Aaker, 1996). Perceived quality is defined as "the consumer's judgment about a product's overall excellence or superiority" (Zeithaml, 1988, p. 3). It is not the objective quality of the product but consumers' subjective evaluations which depend on their perceptions. Aaker (1991, p. 109) defines brand associations as "anything linked to the memory of a brand". These associations can derive from a wide range of sources and vary according to their favorability, strength, and uniqueness (Keller, 1993). Although brand associations can be classified into different types, three main groups related to the product, the personality and the organisation are identified in the literature (Aaker, 1996; Chen, 2001; Pappu *et al.*, 2005). Finally, Aaker (1991, p. 39) defines brand loyalty as "the attachment that a customer has to a brand". Brand loyalty can be conceptualised in several ways, for example based on a behavioural perspective, which emphasizes repeat purchasing behaviour (Ehrenberg *et al.*, 1990) or on an attitudinal perspective, which includes a commitment in terms of some unique values associated with the brand (Chaudhuri and Holbrook, 2001).

Following these frameworks, authors such as Yoo *et al.* (2000), Yoo and Donthu (2001, 2002), Washburn and Plank (2002), Netemeyer *et al.* (2004) and Pappu *et al.* (2005) have developed and empirically tested scales to measure brand equity. Focusing on a consumer perspective, these researchers have drawn on different dimensions of brand equity (e.g. brand awareness, brand associations, perceived quality, brand loyalty, perceived value, brand uniqueness, etc.).

Researchers have identified scope for improving the measurement of brand equity (Yoo *et al.*, 2000; Yoo and Donthu, 2001; Washburn and Plank, 2002; Atilgan *et al.*, 2005). While some researchers have sought to improve the measurement of brand equity (Pappu *et al.*, 2005), there still remains limitations such as the use of a single measure for some brand equity dimensions (e.g. brand awareness), limiting its use in confirmatory factor analysis.

Furthermore, a substantial literature has focused on the conceptualisation and measurement of brand equity within a single country or culture. To our knowledge, only Yoo and Donthu (2001, 2002) have examined the generalisability of their brand equity scale across several cultures (US and Korea) within the cognitive psychology paradigm. Similarly, within the information economics paradigm, only Erdem *et al.* (2006) have tested how their conceptualisation of brand equity explains consumer choice and brand equity formation in different countries.

Therefore, our study tests the cross-national applicability of the brand equity scale in two different countries: the UK and Spain. The proposed scale seeks to improve the measurement of brand equity incorporating different types of brand associations (Yoo and Donthu, 2001) and multiple measures for all the brand equity dimensions (Pappu *et al.*, 2005). Finally we use a sample of consumers (non-students) to test the measurement invariance of our scale unlike previous research (Yoo and Donthu, 2001, 2002).

Measurement invariance issue in cross-national research

To draw valid conclusions regarding the cross-national generalisability of constructs, models or theories, measurement invariance must be established. Otherwise, results can lead to ambiguous or even erroneous conclusions.

Measurement invariance concerns "whether or not, under different conditions of observing and studying phenomena, measurement operations yield measures of the same attribute" (Horn and McArdle, 1992, p. 117). Therefore if measurement invariance is met, it implies that the set of indicators is measuring the same latent variables across groups (Kline, 2005).

Steenkamp and Baumgartner (1998) identified different forms of measurement invariance that should be satisfied

Journal of Product & Brand Management
Volume 17 · Number 6 · 2008 · 384–392

depending on the goal of the study. In this paper we seek to establish whether brand equity can be conceptualised in the same way across countries and to explore whether difference scores on the items can be meaningfully compared across countries. According to these objectives, configural and metric invariance were examined.

Configural invariance exists when the same factor structures are identified across the analysed groups (Steenkamp and Baumgartner, 1998). This criterion is satisfied if the basic model structure (i.e. the pattern of fixed and nonfixed parameters) is invariant across groups. This initial baseline model has no constraints on estimated parameters, permitting different parameter values across countries. This model provides the bases for comparison with the subsequent models.

Metric invariance is the second assumption that must be met to ensure that different groups respond to the items in the same way (Steenkamp and Baumgartner, 1998). This form of measurement invariance has also been referred to as weak factorial invariance (Meredith, 1993). Metric invariance is achieved when all factor loading parameters are equal across groups. Thus, the test of metric invariance requires constraining the factor pattern coefficients to be equal across countries. Because of this, the model with metric invariance is more restrictive than the baseline model.

To examine both tests of equivalence, a well-fitting baseline measurement model is first developed for each group. Then, the configural invariance of the model across groups is tested. In the subsequent stage, if configural invariance has been supported the equivalence of factor loadings across groups on each dimension (metric invariance) is tested.

If both forms of measurement invariance are met, the construct analysed has the same basic meaning and structure cross-nationally and item differences across groups reflect true differences. As such the measurement can be used cross-nationally to establish the generalisability of theories.

Methodology

When conducting cross-national research, inequivalent or biased information can lead to erroneous results. Therefore the equivalence of data collected across countries is regarded as a key issue (Sekaran, 1983; Hui and Triandis, 1985; Singh, 1995; Craig and Douglas, 2000). van Herk *et al.* (2005) identify two approaches to analysing equivalence. Some researchers focus on the research process and the procedures that must be addressed before data collection (e.g. conceptual, functional and category equivalence; translation equivalence; data collection techniques; sampling). Other researchers emphasise the analysis and interpretation of data. Since we want to test the measurement invariance of the brand equity scale, attention has to be paid to both approaches (Singh, 1995).

In the first stage of the research, particular attention needs to be paid to construct equivalence. Briefly, the examination of construct equivalence is concerned with three aspects (Craig and Douglas, 2000). These are functional equivalence, which relates to whether the constructs, objects, etc. being studied serve the same function across countries; conceptual equivalence, which refers to whether the same constructs, stimuli, objects, etc. occur and are expressed in similar ways; and category equivalence, which addresses whether the same classification scheme of objects or other stimuli can be employed across countries.

In establishing functional and conceptual equivalence, the brand equity construct and its dimensions appear to be applicable across the UK and Spain. These concepts (i.e. brand loyalty, brand awareness, perceived quality, brand associations) have the same role and similar interpretation in both countries.

Similarly, construct equivalence was considered when selecting the product categories and brands. The criteria followed were that both product categories and brands were widely available and well-known to UK and Spanish consumers. Furthermore we made sure that they have the same function for consumer in both countries. These guidelines ensure the functional and conceptual equivalence related to these aspects (Craig and Douglas, 2000; Usunier, 2000).

According to these criteria, the product categories and brands were selected using the Best Global Brands 2006 ranking of Interbrand (2006). The use of rankings and secondary research is usual in brand equity research (e.g. Cobb-Walgren *et al.*, 1995; Krishnan, 1996; Netemeyer *et al.*, 2004). Two strong, mature brands in each of four diverse product categories were chosen: Coca-Cola and Pepsi for soft drinks; Adidas and Nike for sportswear; Sony and Panasonic for consumer electronics; and BMW and Volkswagen for cars. The categories reflect a broad set of consumer products to provide some generalisability.

Sample and procedure

To examine the cross-national applicability of this scale, data was collected from two countries: the UK and Spain. Cultural differences exist between the UK and Spain (Hofstede, 1984; Kumar, 2000). According to Hofstede's cultural framework (Hofstede, 1984), the UK has a lower power distance and uncertainty avoidance than Spain, although it is more individualistic and masculine.

Eight different questionnaires were designed, one for each brand. Unlike some of the prior research in brand equity, this research is based on consumers rather than students. Data was collected at several locations in the city of Birmingham in the UK and the city of Zaragoza in Spain. Using quota sampling by age and sex, a similar size and demographic structure was selected in both countries. In total 417 completed questionnaires in the UK and 414 in Spain were collected. Non-valid questionnaires were discarded, resulting in 411 valid questionnaires in each country.

The questionnaire was administered in English in the UK and in Spanish in Spain. A back-translation process (Brislin, 1970; Craig and Douglas, 2000) was employed to ensure the development of comparable versions of the questionnaire in English and Spanish.

Measures

A literature review was undertaken to select the most appropriate way to measure each variable. Recall, recognition and familiarity with the brand were used as measures of brand awareness (Yoo *et al.*, 2000; Netemeyer *et al.*, 2004). Perceived quality was measured using the items proposed by Pappu *et al.* (2005, 2006). Items proposed by Yoo *et al.* (2000) were used to measure brand loyalty as the overall attitudinal loyalty to the brand. To measure brand associations we follow Yoo and Donthu's (2001) suggestion.

Journal of Product & Brand Management
Volume 17 · Number 6 · 2008 · 384–392

They advocate that brand equity dimensions may be expanded to clarify the structure of this construct in detail. Therefore we included three kinds of associations widely recognised in the literature: perceived value, brand personality and organisational associations (Aaker, 1996; Chen, 2001; Pappu *et al.*, 2005). Based on Lassar *et al.* (1995), Aaker (1996), and Netemeyer *et al.* (2004) three items were used to measure perceived value. Brand personality was measured following Aaker (1996). Finally, for organisational brand associations, Aaker's (1996) and Pappu's *et al.* (2005, 2006) proposals were followed. In all cases, seven-point Likert-type questions were used with 1 = strongly disagree and 7 = strongly agree. A list of the items used to measure each construct is provided in Table I.

Model evaluation criteria

To evaluate the hypothesised models and determine whether the fit declines substantially as invariance parameters are imposed, the difference in the Satorra-Bentler scaled chi-square (S-Bχ^2) values was used. If this difference value is statistically significant, it suggests that the constraints specified in the more restrictive model do not hold. By contrast, if this difference value is statistically non-significant, it suggests that the specified equality constraints are tenable. In addition, since researchers have argued that this difference in the S-Bχ^2 value (ΔS-Bχ^2) is sensitive to sample size, two more alternative criteria were employed. First, an adequate fit of the model to the data. We evaluated the models using the normed-fit index (NFI), the non-normed fit index (NNFI), the comparative fit index (CFI), the incremental fit index (IFI) and the root mean square error of approximation (RMSEA). Values of 0.9 or above for the NFI, NNFI, CFI and IFI and values of RMSEA of 0.08 or smaller indicate a good fit (Hair *et al.*, 2006). Second, a negligible ΔCFI value between the models of -0.01 or less (Cheung and Rensvold, 2002; Byrne, 2006). Finally, identification of missfitting parameters (e.g. equality constraints non-invariant across groups) was accomplished by means of the lagrange multiplier test (LM). In the LM test, the viability of fixed parameters is assessed univariately and multivariately to determine those parameters that would significantly decrease the χ^2 if they were freely estimated in a future run (Byrne, 2006).

Results

Measurement invariance was assessed using multi-group confirmatory factor analysis, since it is considered the most appropriate approach for evaluating measurement scales

Table I Brand equity items

Constructs and measurement	Brand awareness (Yoo *et al.*, 2000; Netemeyer *et al.*, 2004)	
	AW1	I am aware of brand X
	AW2	When I think of PC, brand X is one of the brands that comes to mind
	AW3	X is a brand of PC I am very familiar with
	AW4	I know what brand X looks like
	AW5	I can recognise brand X amongst other competing brands of PC
	Perceived quality (Pappu *et al.*, 2005, 2006)	
	PQ1	Brand X offers very good quality products
	PQ2	Brand X offers products of consistent quality
	PQ3	Brand X offers very reliable products
	PQ4	Brand X offers products with excellent features
	Brand loyalty (Yoo *et al.*, 2000)	
	LO1	I consider myself to be loyal to brand X
	LO2	Brand X would be my first choice when considering PC
	LO3	I will not buy other brands of PC if brand X is available at the store
Brand associations	Perceived value (Lassar *et al.*, 1995; Aaker, 1996; Netemeyer *et al.*, 2004)	
	AS1	Brand X is good value for money
	AS2	Within PC I consider brand X a good buy
	AS3	Considering what I would pay for brand X, I would get much more than my money's worth
	Brand personality (Aaker, 1996)	
	AS4	Brand X has a personality
	AS5	Brand X is interesting
	AS6	I have a clear image of the type of person who would use brand X
	Organisational associations (Aaker, 1996; Pappu *et al.*, 2005, 2006)	
	AS7	I trust the company which makes brand X
	AS8	I like the company which makes brand X
	AS9	The company which makes brand X has credibility

Note: PC – product category

cross-nationally (Steenkamp and Baumgartner, 1998). However, before reporting the multi-group CFA results, the items were evaluated using exploratory techniques to assess reliability and dimensionality, followed by a confirmatory factor analysis for each country. In this way the baseline measurement models in each country were established.

Baseline measurement models in each country

First we explored whether a similar pattern of internal consistency and dimensionality among brand equity dimensions was found in each country sample. To assess the reliability of our measures we used Cronbach's alpha and the item-total correlation. We then performed exploratory factor analyses using principal components analysis with varimax rotation. Results suggested that the scales were highly reliable and that the corresponding items of each scale clustered into a single factor with significant factor loadings. The explained variance exceeded 60 per cent in each case. Only the items of brand associations dimension loaded on more than one factor.

As earlier explained, brand associations was characterised by three types of associations (perceived value, brand personality and organisational associations). To examine the dimensionality of brand associations in more detail, we established and tested different sets of measurement models (one-, two- and three-dimensional) using confirmatory factor analysis in both countries. After comparing them using the S-Bχ^2 difference tests and other values of fit (e.g. NFI, NNFI, CFI, IFI and RMSEA) and analysing the psychometric properties in each case, results suggested that a three-dimensional model in both countries was the best model. The standardised loadings of the items to their respective constructs were significant with values above 0.5, suggesting the convergent validity of each factor. Composite reliability (CR) and average variance extracted (AVE) indices exceeded the recommended minimum standard of 0.6. Discriminant validity was also supported in both countries since none of the confidence intervals around the correlation estimate between any two factors contained a value of one.

Therefore, the first brand association dimension (items AS1, AS2 and AS3) refers to the product associations (perceived value). The second dimension (items AS4 and AS5; item AS6 was deleted in both samples since the R^2 was below 0.4) refers to the brand personality. The third dimension (items AS7, AS8 and AS9) is related to organisational associations.

Some researchers have advocated studying brand associations individually to better guide brand decisions (del Río *et al.*, 2001). As such, according to the results of brand associations' dimensionality, in the following analyses, these dimensions were included as three different dimensions.

A confirmatory factor analysis in each sample was then performed using EQS 6.1 and the robust maximum-likelihood estimation (Bentler, 1995). Results of the estimation of these six-factor measurement models indicated that the constructs were measured adequately through their indicators in both the UK and Spain. Tables II and III detail the results of the measurement models.

All factor loadings were highly significant with their standardised estimates ranging from 0.732 to 0.932 in the UK and from 0.668 to 0.939 in Spain. Likewise, the coefficients had a clear relation with the underlying factors ($R^2 > 0.4$). Composite reliability (CR) and average variance extracted (AVE) indices of each construct were above the recommended minimum of 0.6 (Bagozzi and Yi, 1988). Tests of the discriminant validity of the six dimensions were also performed for both samples. Again, discriminant validity was supported by the fact that none of the confidence intervals around the correlation estimate between any two factors contained a value of one. Finally, as can be seen in Table IV, tests of goodness of fit indicated an acceptable level (Kline, 2005; Hair *et al.*, 2006).

In addition to these six-factor measurement models in both samples, and similarly to the process followed in the analysis of brand associations dimension, a number of competing models were estimated. However, the initial model turned out to be the best fitting model for both the English and Spanish samples.

In conclusion, taken together these results suggest that the measurement models in both countries appear to fit the data reasonably well.

Table II Measurement analysis results: parameter estimates

	UK		Spain		Common	
Items	Lambda*	*t*-value	Lambda*	*t*-value	Lambda*	*t*-value
AW1	1.000	–	1.000	–	1.000	–
AW2	1.093	19.36	1.423	7.28	1.241	13.81
AW3	1.200	15.52	1.457	7.75	1.332	13.20
AW4	1.173	16.47	1.388	9.97	1.284	16.61
AW5	1.166	15.74	1.202	6.52	1.218	13.61
PQ1	1.000	–	1.000	–	1.000	–
PQ2	0.966	22.02	0.993	45.88	0.982	45.67
PQ3	1.000	28.41	0.965	21.44	0.986	35.73
PQ4	0.942	20.67	1.007	31.52	0.985	37.22
LO1	1.000	–	1.000	–	1.000	–
LO2	1.073	30.18	1.066	29.98	1.068	42.32
LO3	0.910	23.24	0.958	24.39	0.935	33.73
AS1	1.000	–	1.000	–	1.000	–
AS2	1.217	22.90	1.201	23.06	1.209	32.55
AS3	0.947	15.49	0.978	19.17	0.963	24.26
AS4	1.000	–	1.000	–	1.000	–
AS5	1.064	18.74	1.090	16.87	1.079	24.46
AS7	1.000	–	1.000	–	1.000	–
AS8	0.988	25.39	1.015	36.07	1.005	44.06
AS9	0.993	23.74	0.994	33.92	0.993	41.35

Notes: Item AS6 was eliminated in both countries during validation process; *Parameter estimates are unstandardised values

Test of configural invariance

Although the results of group-level exploratory factor analyses and confirmatory factor analyses have provided strong indications of configural invariance, a multi-group confirmatory factor analysis was conducted to test this first level of measurement invariance. Testing for configural invariance requires only that the same number of factors and factor loading pattern be the same across groups. As such no constraints are imposed on the parameters. Thus a multi-group analysis of the baseline models for the UK and Spain was conducted.

The fit of this model was acceptable indicating that the brand equity scale has a consistent factor structure across the two countries. As shown in Table V, the RMSEA was 0.064 (less than 0.08) and the other indices of fit were above 0.9.

Table III Measurement analysis results: reliability and average variance extracted

Factor	UK		Spain	
	Composite reliability	Average variance extracted	Composite reliability	Average variance extracted
Brand awareness	0.899	0.642	0.887	0.612
Perceived quality	0.929	0.765	0.950	0.825
Brand loyalty	0.906	0.763	0.918	0.788
Brand associations: perceived value	0.878	0.708	0.873	0.697
Brand associations: brand personality	0.869	0.769	0.882	0.789
Brand associations: organisation	0.923	0.799	0.953	0.871

Table IV Measurement analysis results: fit indices

Model fit indices	S-Bχ^2 (df, *p*)	NFI	NNFI	CFI	IFI	RMSEA
UK	407.49 (155, 0.000)	0.917	0.935	0.947	0.947	0.064
Spain	415.20 (155, 0.000)	0.888	0.909	0.926	0.927	0.063

Table V Assessment of measurement invariance

Model specification	S-Bχ^2 (df, *p*)	NFI	NNFI	CFI	IFI	RMSEA	ΔS-Bχ^2 *	Δdf	ΔCFI
M1: configural invariance	822.83 (310, 0.000)	0.902	0.922	0.936	0.937	0.064	–	–	–
M2: metric invariance	835.36 (324, 0.000)	0.901	0.925	0.936	0.937	0.063	13.26	14	0.000

Note: *The corrected value is presented since ΔS-Bχ^2 is not distributed as χ^2
Source: Byrne (2006)

On the basis of the criteria of fit indices we conclude that the hypothesised model fits the data well cross-nationally, with significant and nonzero factor loadings on salient factors and zero loadings on non-salient factors. The factor structure for the consumer-based brand equity scale is invariant across the UK and Spain and has configural invariance.

Test of metric invariance

After configural invariance was established, to test for metric invariance the factor pattern coefficients were constrained to be equal. It can be seen from Table V that there was a non-signficant increase in the S-Bχ^2 between the unconstrained model (M1) and the constrained model (M2) (ΔS$-$B$\chi^2 = 13.26$, $\Delta df = 14$, $p > 0.05$). Similarly the CFI values are the same in both models and the alternative fit indices indicated an acceptable fit. Examination of the LM test did not reveal parameters operating non-equivalently across countries. These results suggested that metric invariance of the brand equity scale was supported across the two countries. The common measurement parameters estimates (unstandardised) for the English and the Spanish samples are provided in Table II.

To ensure that these results are accurate and not affected by the indicators selected to fix its loading to 1.0 (requirement for identification), the model was reanalysed after fixing the loadings of other indicators to 1.0 (Cheung and Rensvold, 1999; Kline, 2005). Results suggested that the factor loadings originally fixed and consequently not tested for group differences were comparable and measurement invariants. As such the factor structure can be considered invariant across the two countries and cross-national consumer responses to the brand equity scale can be meaningfully compared.

Discussion and implications

Due to the competitive advantages provided by brands, it has become vital for brand managers to have valid and reliable instruments to measure brand equity (Pappu *et al.*, 2005). In addition, the internationalisation of companies has increased the need for conducting research beyond domestic markets with instruments that have cross-national equivalence.

Measurement invariance is an important issue in cross-national research (Steenkamp and Baumgartner, 1998). However, in the literature measurement invariance is not always thoroughly addressed. Researchers often assume that an instrument devised in one country is universally applicable.

Despite the notable attention paid to the brand equity construct, little research has examined its cross-national applicability. In this study we assessed the measurement invariance of the consumer-based brand equity scale in UK and Spain. We tested whether brand equity can be conceptualised in the same way across countries and whether difference scores on the items can be meaningfully compared among these countries. Therefore, configural and metric invariance were analysed. We also addressed some of the limitations that other brand equity scales have. Our study used a sample of consumers (non-students) unlike other studies (Yoo *et al.*, 2000; Yoo and Donthu, 2001, 2002; Atilgan *et al.*, 2005) and incorporated different types of brand associations (Yoo and Donthu, 2001) and multiple measures for the brand equity dimensions (Pappu *et al.*, 2005).

The results showed that the factor structure of our brand equity scale was invariant across the two countries (configural invariance). Furthermore, it was found that cross-national consumer responses to brand equity scale can be meaningfully compared (metric invariance), indicating that consumers in both countries interpret and respond to the items in an equivalent manner.

These results have important implications for researchers and the management of international companies. As the brand equity scale is invariant across countries, researchers and managers can use this instrument to measure and manage brand equity across the UK and Spain. This should improve the reliability, validity and comparability of research findings.

A key concern of academics and managers is to measure concepts. Knowing the extent to which the consumer-based brand equity scale is generalisable across countries is important since it will allow managers to design and implement efficient brand strategies in international contexts. The use of this instrument should contribute to better decisions about building and managing brand equity.

Similarly, this measurement may be suitable for testing theoretical and conceptual relationships. For instance, the cross-national brand equity scale can be used to examine its antecedents and consequences in several countries.

There are several limitations with our study that suggest directions for further research. Given that the study was conducted in the UK and Spain, the brand equity scale is strictly generalisable across these two countries. Future research should examine whether the scale is cross-national invariant in other contexts. Research between countries and continents will contribute to the development of a more generalisable brand equity scale. Further research should also examine the brand equity scale among other products and services in different pan country contexts.

References

Aaker, D.A. (1991), *Managing Brand Equity: Capitalizing on the Value of Brand Name*, The Free Press, New York, NY.

Aaker, D.A. (1996), "Measuring brand equity across products and markets", *California Management Review*, Vol. 38 No. 3, pp. 102-20.

Ailawadi, K.L., Lehmann, D.R. and Neslin, S.A. (2003), "Revenue premium as an outcome measure of brand equity", *Journal of Marketing*, Vol. 67, pp. 1-17.

Ambler, T. (2003), *Marketing and the Bottom Line*, 2nd ed., FT Prentice-Hall, London.

Atilgan, E., Aksoy, S. and Akinci, S. (2005), "Determinants of the brand equity: a verification approach in the beverage industry in Turkey", *Marketing Intelligence & Planning*, Vol. 23 No. 3, pp. 237-48.

Bagozzi, R.P. and Yi, Y. (1988), "On the evaluation of structural equation models", *Journal of Academy of Marketing Science*, Vol. 16 No. 1, pp. 74-94.

Bentler, P.M. (1995), *EQS Structural Equations Program Manual*, Multivariate Software Inc., Encino, CA.

Brislin, R.W. (1970), "Back-translation for cross-cultural research", *Journal of Cross-Cultural Psychology*, Vol. 1 No. 3, pp. 185-216.

Byrne, B.M. (2006), *Structural Equation Modeling with EQS: Basic Concepts, Applications, and Programming*, 2nd ed., Lawrence Erlbaum Associates, Inc., Mahwah, NJ.

Chaudhuri, A. and Holbrook, M.B. (2001), "The chain of effects from brand trust and brand affect to brand performance: the role of brand loyalty", *Journal of Marketing*, Vol. 65, April, pp. 81-93.

Chen, K.J. (2001), "Using free association to examine the relationship between the characteristics of brand associations and brand equity", *Journal of Product & Brand Management*, Vol. 10 No. 7, pp. 439-51.

Cheung, G.W. and Rensvold, R.B. (1999), "Testing factorial invariance across groups: a reconceptualization and proposed new method", *Journal of Management*, Vol. 25 No. 1, pp. 1-27.

Cheung, G.W. and Rensvold, R.B. (2002), "Evaluating goodness-of-fit indices for testing measurement invariance", *Structural Equation Modeling*, Vol. 9 No. 2, pp. 233-55.

Cobb-Walgren, C., Ruble, C.A. and Donthu, N. (1995), "Brand equity, brand preference and purchase intent", *Journal of Advertising*, Vol. 24 No. 3, pp. 25-40.

Craig, C.S. and Douglas, S.P. (2000), *International Marketing Research*, 2nd ed., John Wiley & Sons, Chichester.

del Río, A.B., Vázquez, R. and Iglesias, V. (2001), "The effects of brand associations on consumer response", *Journal of Consumer Marketing*, Vol. 18 No. 5, pp. 410-25.

Durvasula, S., Andrews, J.C., Lysonki, S. and Netemeyer, R.G. (1993), "Assessing the cross-national applicability of consumer behaviour models: a model of attitude toward advertising in general", *Journal of Consumer Research*, Vol. 19 No. 4, pp. 626-36.

Ehrenberg, A.S.C., Goodhardt, G.J. and Barwise, P.B. (1990), "Double jeopardy revisited", *Journal of Marketing*, Vol. 54, pp. 82-91.

Erdem, T., Swait, J. and Valenzuela, A. (2006), "Brands as signals: a cross-country validation study", *Journal of Marketing*, Vol. 70, January, pp. 34-49.

Hair, J.F., Black, B., Babin, B., Anderson, R.E. and Tatham, R.L. (2006), *Multivariate Data Analysis*, 6th ed., Prentice-Hall, Upper Saddle River, NJ.

Hofstede, G. (1984), *Culture's Consequences: International Differences in Work-Related Values*, Sage Publications, Newbury Park, CA.

Horn, J.L. and McArdle, J. (1992), "A practical and theoretical guide to measurement invariance in aging research", *Experimental Aging Research*, Vol. 18, pp. 117-44.

Hui, C.H. and Triandis, H.C. (1985), "Measurement in cross-cultural psychology: a review and comparison of strategies", *Journal of Cross-Cultural Psychology*, Vol. 16 No. 2, pp. 131-52.

Interbrand (2006), "Best global brands: a ranking by brand value", available at: www.interbrand.com/best_brands_2006.asp

Keller, K.L. (1993), "Conceptualizing, measuring, and managing customer-based brand equity", *Journal of Marketing*, Vol. 57, January, pp. 1-22.

Keller, K.L. (2007), *Strategic Brand Management: Building, Measuring, and Managing Brand Equity*, 3rd ed., Prentice-Hall, New York, NY.

Kline, R.B. (2005), *Principles and Practice of Structural Equation Modeling*, 2nd ed., The Guilford Press, New York, NY.

Knight, G.A., Spreng, R.A. and Yaprak, A. (2003), "Cross-national development and validation of an international business measurement scale: the COISCALE", *International Business Review*, Vol. 12, pp. 581-99.

Krishnan, H.S. (1996), "Characteristics of memory associations: a consumer-based brand equity perspective", *International Journal of Research in Marketing*, Vol. 13, pp. 389-405.

Kumar, V. (2000), *International Marketing Research*, Prentice-Hall, Upper Saddle River, NJ.

Lassar, W., Mittal, B. and Sharma, A. (1995), "Measuring customer-based brand equity", *Journal of Consumer Marketing*, Vol. 12 No. 4, pp. 11-19.

Mavondo, F., Gabbott, M. and Tsarenko, Y. (2003), "Measurement invariance of marketing instruments: an implication across countries", *Journal of Marketing Management*, Vol. 19, pp. 523-40.

Meredith, W. (1993), "Measurement invariance, factor analysis, and factorial invariance", *Psychometrika*, Vol. 58, December, pp. 525-43.

Mullen, M.R. (1995), "Diagnosing measurement equivalence in cross-national research", *Journal of International Business Studies*, Vol. 26 No. 3, pp. 573-96.

Netemeyer, R., Krishnan, B., Pullig, C., Wang, G., Yagci, M., Dean, D., Ricks, J. and Wirth, F. (2004), "Developing and validating measures of facets of customer-based brand equity", *Journal of Business Research*, Vol. 57, pp. 209-24.

Pappu, R., Quester, P.G. and Cooksey, R.W. (2005), "Consumer-based brand equity: improving the measurement empirical evidence", *Journal of Product & Brand Management*, Vol. 14 No. 3, pp. 143-54.

Pappu, R., Quester, P.G. and Cooksey, R.W. (2006), "Consumer-based brand equity and country-of-origin relationships", *European Journal of Marketing*, Vol. 40 Nos 5/6, pp. 696-717.

Parameswaran, R. and Yaprak, A. (1987), "A cross-national comparison of consumer research measures", *Journal of International Business Studies*, Vol. 18, Spring, pp. 35-49.

Rangaswamy, A., Burke, R. and Oliva, T.A. (1993), "Brand equity and the extendibility of brand names", *International Journal of Research in Marketing*, Vol. 10, pp. 61-75.

Sekaran, U. (1983), "Methodological and theoretical issues and advancements in cross-cultural research", *Journal of International Business Studies*, Vol. 14, pp. 61-73.

Singh, J. (1995), "Measurement issues in cross-national research", *Journal of International Business Studies*, Vol. 26 No. 3, pp. 593-619.

Steenkamp, J.E.M. and Baumgartner, H. (1998), "Assessing measurement invariance in cross-national consumer research", *Journal of Consumer Research*, Vol. 25, pp. 78-90.

Usunier, J.C. (2000), *Marketing across Cultures*, 3rd ed., Pearson Education Limited, Harlow.

van Herk, H., Poortinga, Y.H. and Verhallen, T.M.M. (2005), "Equivalence of survey data: relevance for international marketing", *European Journal of Marketing*, Vol. 39 Nos 3/4, pp. 351-64.

Vandenberg, R.J. and Lance, C.E. (2000), "A review and synthesis of measurement invariance literature: suggestions, practices, and recommendations for organisational research", *Organisational Research Methods*, Vol. 3 No. 1, pp. 4-69.

Washburn, J.H. and Plank, R.E. (2002), "Measuring brand equity: an evaluation of a consumer-based brand equity scale", *Journal of Marketing Theory and Practice*, Vol. 10 No. 1, pp. 46-62.

Yoo, B. and Donthu, N. (2001), "Developing and validating a multidimensional consumer-based brand equity scale", *Journal of Business Research*, Vol. 52, pp. 1-14.

Yoo, B. and Donthu, N. (2002), "Testing cross-cultural invariance of the brand equity creation process", *Journal of Product & Brand Management*, Vol. 11 No. 6, pp. 380-98.

Yoo, B., Donthu, N. and Lee, S. (2000), "An examination of selected marketing mix elements and brand equity", *Journal of the Academy of Marketing Science*, Vol. 28 No. 2, pp. 195-211.

Zeithaml, V.A. (1988), "Consumer perceptions of price, quality, and value: a means-end model and synthesis of evidence", *Journal of Marketing*, Vol. 52, July, pp. 2-22.

About the authors

Isabel Buil is a PhD candidate in Marketing and an Assistant Professor in the Department of Economy and Business Studies at University of Zaragoza, Spain. She has been Visiting Scholar at the Centre for Research in Brand Marketing at the Birmingham Business School, England. She is a member of the research group Generes recognized by the Government of Aragon. She has published research papers in both local and international journals, such as the *Business Strategy and Environment*. She has attended and presented research papers to several national and international conferences, such as EMAC Conference. Isabel Buil is the corresponding author and can be contacted at: ibuil@unizar.es

Leslie de Chernatony is Professor of Brand Marketing and Director of the Centre for Research in Brand Marketing at Birmingham University Business School. With a doctorate in brand marketing, he has a substantial number of publications in American and European journals and is a regular presenter at international conferences. He has several books on brand marketing, the two most recent being *Creating Powerful Brands and From Brand Vision to Brand Evaluation*. A winner of several research grants, his two most recent grants have supported research into factors associated with high performance brands and research into services branding. He was Visiting Professor at Madrid Business School and is currently Visiting Professor at Thammasat University, Bangkok and University of Lugano, Switzerland. Leslie is a Fellow of the Chartered Institute of Marketing and Fellow of the Market Research Society. He acts as an international consultant to organisations seeking more effective brand strategies and has run acclaimed branding seminars throughout Europe, Asia, America and the Far East. He is an experienced expert witness in legal cases involving branding issues in commercial and competition cases.

Eva Martínez is a PhD in Business and Economics Studies and Professor in Marketing in the Department of Economy and Business Studies at Faculty of Economics and Business, University of Zaragoza, Spain. She has been Visiting Scholar at the Centre for Research in Brand Marketing at the Birmingham Business School, England. She has won several research grants from the Spanish Ministry of Education and Culture. Eva is a member of the research group Generes recognised by the Government of Aragon. She is a frequent presenter at national and international conferences. She has cooperated in several international books. She has published several research papers in international journals such as the *Journal of Consumer Marketing*, *Journal of Product and Brand Management*, *European Journal of Marketing*, *Journal of Business Research*, *Journal of Marketing Management* and *International Marketing Review.*

Journal of Product & Brand Management
Volume 17 · Number 6 · 2008 · 384–392

Executive summary and implications for managers and executives

This summary has been provided to allow managers and executives a rapid appreciation of the content of the article. Those with a particular interest in the topic covered may then read the article in toto *to take advantage of the more comprehensive description of the research undertaken and its results to get the full benefit of the material present.*

A focus on measuring brand equity across borders

Being able to measure brand equity has become an increasingly significant pre-occupation for marketers. The case for the importance of building brand equity is proven. It provides the platform from which companies can seriously capitalise in terms of higher margins, opportunities to create brand extensions, improved customer communications, higher customer preferences and enhanced purchasing intentions. With the importance of building brand equity being increasingly realised then to wish to measure it effectively is a natural progression.

It is an area where there is room for further development, particularly looking for measurements that stand up across international boundaries – country managers may thrive on country-specific information but brand managers need to paint on a bigger canvas. Research by Isabel Buil, Leslie de Chernatony and Eva Martinez of the University of Zaragoza in Spain, and University of Birmingham in the UK sought to validate a scale for measuring consumer-based brand equity.

Unsurprisingly their study was conducted in the UK and Spain, specifically in Birmingham and Zaragoza. The survey was of actual consumers, not the captive audience of students, which provides a broader base from which to analyse the findings.

The first challenge for the research team was to ascertain whether brand equity can be thought about in the same way in different countries. The second challenge to discover whether different scores on a scale could be meaningfully compared. It seems like they can. Certainly the key concepts of brand loyalty, brand awareness, perceived quality and brand associations appear to have the same role and are interpreted in the same way in both countries.

Brand equity is meaningful, it can be measured, and it can be measured internationally. The prize for examining these issues is the ability to create and implement an efficient international brand strategy.

Big brands in the mix in the quest to generalize

To test their model effectively, four diverse consumer brand categories were chosen and two mature brands analysed within each. These were:

1 Soft drinks – Coca-Cola and Pepsi.
2 Sportswear – Adidas and Nike.
3 Consumer electronics – Sony and Panasonic.
4 Cars – BMW and Volkswagen.

Arguably there could hardly be bigger brands in the mix, although the researchers' rationale in choosing such broad categories was to achieve that great god of "generalisability". Research needs to be specific, but to be useful findings need to be said to generally apply. Cultural considerations were also considered, represented by Hostede's cultural dimensions, given the need to generalise across borders.

A key finding was that consumers in both countries responded to items of brand equity in the same way, which must surely be good news for international marketers.

What to make of these outcomes

With the internationalisation of business, and there can be few more powerful ways of demonstrating its progression than listing these eight brands, the international development of brand equity metrics needs to step up a-pace. The case for measuring brand equity does not need revisiting. The case is proven. It is one that would not appear to need revisiting, albeit no doubt someone, somewhere soon will.

The case would now seem to be proven too on the potential to use the same metrics on an international basis. Life can be complicated enough. This finding would seem to make it that little bit simpler. This has been an under-researched area and lazy assumptions that measurements developed in one country will universally apply cry out for this type of robust testing.

Brand equity scales have existed before, but this study moves things along and addresses earlier shortcomings. However, its main benefit is in establishing that consumers in two different countries think about brands in the same way and that measures of brand equity can be directly compared using one scale. It is a scale that brand managers can pick up and use. It has been shown to be effective in two countries and, rather like Web 2.0, will get better the more it is used. Usage will help improve the reliability, validity and comparability of the research findings. As such, this work represents a new start, or possibly a launch pad.

Measurement plays a key role in driving marketing strategy. When skies are blue then intuition may rule. When the rubber hits the road hard numbers win arguments. The scale created and tested here could form the basis from which brand managers can create and implement efficient international brand management strategies. It can provide the data from which to make better, more informed decisions.

Reading the wind or examining the tealeaves may give comfort, but effective metrics are more likely to help managers make the key decisions and get the major calls right. Judgement in brand management is vital, but it is nice to have an opportunity to base decisions on firm foundations.

(A précis of the article "A cross-national validation of the consumer-based brand equity scale". Supplied by Marketing Consultants for Emerald.)

This article has been cited by:

1. Musa Pinar, Paul Trapp, Tulay Girard, Thomas E. Boyt. 2014. University brand equity: an empirical investigation of its dimensions. *International Journal of Educational Management* **28**:6, 616-634. [Abstract] [Full Text] [PDF]
2. J. Augusto Felício, Margarida Duarte, Vítor Caldeirinha, Ricardo Rodrigues. 2014. Franchisee-based brand equity and performance. *The Service Industries Journal* **34**, 757-771. [CrossRef]
3. Luming Wang, Adam Finn. 2014. A cross-product category CBBE study: item response theory perspective. *Journal of Product & Brand Management* **23**:3, 200-206. [Abstract] [Full Text] [PDF]
4. Marc Fetscherin, Michèle Boulanger, Cid Gonçalves Filho, Gustavo Quiroga Souki. 2014. The effect of product category on consumer brand relationships. *Journal of Product & Brand Management* **23**:2, 78-89. [Abstract] [Full Text] [PDF]
5. Mahdi Rajabi, Nathalie Dens, Patrick De PelsmackerAttention and Memory Effects of Advertising in an International Context 313-337. [CrossRef]
6. Henry Tsai, Ada Lo, Catherine Cheung. 2013. Measuring Customer-Based Casino Brand Equity and its Consequences. *Journal of Travel & Tourism Marketing* **30**, 806-824. [CrossRef]
7. Isabel Buil, Eva Martínez, Leslie de Chernatony. 2013. The influence of brand equity on consumer responses. *Journal of Consumer Marketing* **30**:1, 62-74. [Abstract] [Full Text] [PDF]
8. Jenni Romaniuk, Magda Nenycz-Thiel. 2013. Behavioral brand loyalty and consumer brand associations. *Journal of Business Research* **66**, 67-72. [CrossRef]
9. Francisco Guzmán, Oriol Iglesias, Ulla Hakala, Johan Svensson, Zsuzsanna Vincze. 2012. Consumer-based brand equity and top-of-mind awareness: a cross-country analysis. *Journal of Product & Brand Management* **21**:6, 439-451. [Abstract] [Full Text] [PDF]
10. Phumisak Smutkupt, Donyaprueth Krairit, Do Ba Khang. 2012. Mobile marketing and consumer perceptions of brand equity. *Asia Pacific Journal of Marketing and Logistics* **24**:4, 539-560. [Abstract] [Full Text] [PDF]
11. Musa Pinar, Tulay Girard, Zeliha Eser. 2012. Consumer-based brand equity in banking industry. *International Journal of Bank Marketing* **30**:5, 359-375. [Abstract] [Full Text] [PDF]
12. Sajjad Ahmad, Muhammad Mohsin Butt. 2012. Can after sale service generate brand equity?. *Marketing Intelligence & Planning* **30**:3, 307-323. [Abstract] [Full Text] [PDF]
13. Leonardo Dioko, Rich Harrill, Felicitas Evangelista, Leonardo A.N. Dioko. 2011. Interpersonal influence and destination brand equity perceptions. *International Journal of Culture, Tourism and Hospitality Research* **5**:3, 316-328. [Abstract] [Full Text] [PDF]
14. Rosa E. Rios, Hernan E. Riquelme. 2010. Sources of brand equity for online companies. *Journal of Research in Interactive Marketing* **4**:3, 214-240. [Abstract] [Full Text] [PDF]
15. Kevin Kam Fung So, Ceridwyn King. 2010. "When experience matters": building and measuring hotel brand equity. *International Journal of Contemporary Hospitality Management* **22**:5, 589-608. [Abstract] [Full Text] [PDF]
16. Byeong-Joon Moon, Wonkyu Park, Sang-Chul Choi. 2010. Relationships among Brand Equity Components: An Exploratory Study of the Moderating Role of Product Type. *Journal of Global Academy of Marketing Science* **20**, 98-108. [CrossRef]
17. Salih Yıldız, Hüseyin Sabri KurtulduFactors Affecting Electronic Service Brand Equity 434-492. [CrossRef]
18. Kijpokin KasemsapThe Role of Brand Management in Emerging Markets 167-184. [CrossRef]